

RUENENBERG, Switzerland

WMO: 066450

Lat: 47.4344N

Lon: 7.8794E

Elev: 610

StdP: 94.21

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -7.0	(d) -12.5	(e) 1.4	(f) -6.8	(g) -10.0	(h) 1.7	(i) -5.3	(j) 13.9	(k) 7.0	(l) 11.5	(m) 6.8	(n) 2.4	(o) 70	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.4	(c) 29.3	(d) 19.2	(e) 27.3	(f) 18.6	(g) 25.6	(h) 18.0	(i) 20.2	(j) 27.0	(k) 19.3	(l) 25.8	(m) 18.5	(n) 24.5	(o) 2.3	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.8	23.3	17.0	13.0	22.1	16.2	12.4	21.4	60.7	27.1	57.7	25.9	55.0	24.6	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.8	(b) 7.4	(c) 5.9	(e) -10.6	(f) 32.0	(g) 2.9	(h) 1.7	(i) -12.7	(j) 33.2	(k) -14.4	(l) 34.2	(m) -16.0	(n) 35.1	(o) -18.2	(p) 36.3		
(4) 9.8	(5) 7.4	(6) 5.9	(7) -11.1	(8) 22.0	(9) 3.1	(10) 0.8	(11) -13.3	(12) 22.5	(13) -15.1	(14) 23.0	(15) -16.9	(16) 23.5	(17) -19.1	(18) 24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	1.0	1.6	5.7	9.8	13.1	17.3	19.0	18.5	14.5	10.7	5.3	1.9
	DBStd	7.61	4.60	4.79	4.48	4.29	3.98	3.96	3.57	3.68	3.50	4.03	3.94	4.18
	HDD10.0	1177	280	235	145	57	15	1	0	0	4	42	147	251
	HDD18.3	3228	538	468	393	258	168	67	36	43	122	238	390	508
	CDD10.0	1145	1	1	11	49	111	219	279	264	141	63	7	1
	CDD18.3	155	0	0	0	1	6	35	57	49	8	1	0	0
	CDH23.3	995	0	0	0	2	31	224	383	327	27	1	0	0
	CDH26.7	239	0	0	0	0	3	49	99	88	1	0	0	0
Wind	WSAvg	2.3	2.8	2.8	2.8	2.3	2.0	1.9	1.8	1.7	1.8	2.0	2.3	2.9
Precipitation	PrecAvg	1093	73	69	70	82	106	111	117	110	88	87	88	93
	PrecMax	1383	187	168	248	179	183	230	204	213	157	163	211	199
	PrecMin	752	16	23	24	9	34	26	37	17	42	11	4	0
	PrecStd	163	44	37	49	45	44	47	37	46	36	41	50	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.8	18.3	23.0	26.8	30.5	31.8	32.6	25.9	21.6	16.1	12.5
		MCWB	8.5	7.2	10.6	13.6	17.5	20.3	19.8	18.8	17.4	14.7	11.3	8.3
	2%	DB	10.7	11.4	15.9	20.6	23.6	28.1	29.3	29.0	23.9	19.4	14.1	10.6
		MCWB	7.4	6.7	8.8	12.1	15.7	19.1	18.8	19.3	17.2	14.1	10.2	7.6
	5%	DB	9.2	9.6	14.0	18.7	21.8	26.0	27.2	26.7	21.9	17.6	12.6	9.4
		MCWB	6.3	5.9	8.0	11.4	14.6	18.4	18.4	18.7	16.1	13.3	9.5	6.8
	10%	DB	7.2	8.0	12.2	16.7	19.8	24.0	25.3	24.7	20.1	16.1	10.9	8.0
		MCWB	4.9	4.8	7.4	10.3	13.7	17.5	17.8	18.0	15.4	12.7	8.5	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	8.4	11.2	14.2	18.1	21.6	21.1	21.1	19.0	15.9	12.0	9.2
		MCDB	11.8	11.5	16.7	21.3	24.8	28.5	28.7	27.9	23.8	19.8	14.6	11.2
	2%	WB	7.9	7.2	9.8	12.9	16.7	20.1	19.9	19.9	17.7	14.7	10.8	8.1
		MCDB	10.1	10.6	14.3	19.2	22.3	26.5	26.9	26.8	22.5	18.0	13.4	10.3
	5%	WB	6.5	6.2	8.8	11.8	15.5	19.0	19.0	19.1	16.7	13.9	9.7	7.1
		MCDB	8.7	9.5	12.8	17.5	20.3	24.8	25.7	25.7	20.8	16.9	12.2	9.2
	10%	WB	5.1	5.1	7.8	10.7	14.2	17.9	18.2	18.3	15.8	13.0	8.6	5.9
		MCDB	7.1	7.8	11.7	16.1	18.8	23.1	24.3	24.0	19.4	15.7	10.7	7.9
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.3	6.9	7.9	7.9	8.2	8.4	7.9	7.1	6.0	4.8	4.5
		MCDBR	6.3	7.6	9.4	10.7	10.5	10.7	11.1	10.8	9.5	8.0	7.1	6.3
		MCWBR	4.2	4.8	5.1	5.7	5.5	5.1	4.8	4.8	4.8	4.3	4.4	4.3
	5% WB	MCDBR	6.0	7.2	8.3	9.8	9.5	10.1	10.0	9.9	8.4	7.3	6.6	5.6
		MCWBR	4.3	4.8	4.8	5.5	5.2	5.1	4.7	4.7	4.5	4.3	4.4	4.2
Clear-Sky Solar Irradiance	taub	0.292	0.316	0.349	0.381	0.392	0.395	0.396	0.389	0.367	0.346	0.313	0.289	
	taud	2.465	2.398	2.353	2.289	2.298	2.326	2.315	2.350	2.394	2.443	2.484	2.469	
	Ebn at Noon	801	849	869	871	872	869	862	853	840	802	765	758	
	Edh at Noon	65	85	104	122	126	123	123	114	99	81	64	58	
All-Sky Solar Radiation	RadAvg	1.12	1.95	3.17	4.34	5.02	5.87	5.64	4.91	3.71	2.33	1.26	0.91	
	RadStd	0.14	0.29	0.47	0.64	0.50	0.54	0.51	0.48	0.36	0.23	0.15	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.31	N/A	N/A	N/A	N/A	N/A
(47) Regional (117 neighbors)		N/A	N/A	N/A	+0.74	+0.41	+0.32	N/A	N/A	+55	+14

Nomenclature: See separate page